Robert Forte

Senior Data Scientist

robert.forte.ds@gmail.com (404) 382-0630 Naples, FL LinkedIn

PROFESSIONAL EXPERIENCE

Senior Data Scientist, *Lily AI* 11/2022 – present

- **Developed and Deployed a Proprietary Large Language Model (LLM):** Led the development of a custom LLM tailored for retail applications, utilizing PyTorch and Hugging Face Transformers. Trained on over 10 million product data points from major retailers, enhancing product tagging accuracy by 15% and improving personalization in e-commerce.
- **Implemented Advanced Image Processing and Classification Models:** Designed and deployed image classification pipelines using OpenCV, PyTorch, and FastAI, processing over 500,000 product images. This initiative improved product recognition accuracy by 20%, significantly reducing manual tagging efforts by 30%.
- **Applied Transformer-Based NLP for Product Description Generation:** Built and fine-tuned a BERT-based NLP model to analyze and enrich product metadata. Automated the creation of SEO-optimized product descriptions, reducing content generation time by 40% and boosting customer engagement metrics by 25%.
- **Developed and Launched the “Product Copy” Feature:** Spearheaded the development of “Product Copy,” an AI-driven content generation tool leveraging LLM-powered insights. This tool automated product title and description creation, leading to a 35% reduction in time-to-market and a 10% increase in organic search traffic.
- **Optimized AI Models Through Data-Driven Insights:** Conducted extensive data analysis using SQL, Pandas, and Scikit-learn, refining AI models based on real-time product performance metrics. Achieved a 12% improvement in product discovery rates and a 15% increase in conversion rates, directly impacting sales growth.

Machine Learning Engineer, *Infinite Sky* 01/2020 – 10/2022

- **Developed AI Models for Athlete Performance Analysis:** Designed and implemented deep learning models using PyTorch to analyze and predict athlete performance metrics. Optimized model architectures to enhance predictive accuracy by 20%, focusing on key biomechanical and statistical factors derived from MLB video data.
- **Built and Managed Large-Scale Data Pipelines on AWS:** Developed and maintained scalable data pipelines utilizing AWS ECS, Aurora, Lambda, and Batch, all managed through AWS CDK. This infrastructure processed terabytes of MLB video data, enabling high-speed feature extraction and model inference for performance analysis.
- **Conducted Advanced Data Analysis and Research:** Applied computer vision and statistical methods to extract key insights from MLB video datasets, leveraging Pandas, NumPy, OpenCV, and Matplotlib. These insights played a crucial role in optimizing model training and improving prediction consistency.
- **Automated Data Processing and Feature Engineering Workflows:** Engineered automated pipelines for data ingestion, preprocessing, and storage using AWS Lambda and Batch, significantly reducing data processing latency by 30%. This enabled real-time analytics for athlete performance tracking.
- **Collaborated with Cross-Functional Teams to Deliver AI-Powered Insights:** Worked closely with data scientists, sports analysts, and software engineers to integrate ML-driven insights into the AI platform. This collaboration resulted in enhanced real-time performance tracking tools tailored for MLB athletes and coaches.

Data Scientist, *Reddit* 08/2017 – 11/2019

- **Contributed to Customer Segmentation Initiatives:** Assisted in building clustering models and behavioral analytics pipelines using Python, Scikit-learn, and SQL to segment Reddit users based on engagement patterns. Helped refine marketing strategies, leading to a 15% improvement in targeted campaign performance.
- **Developed Data-Driven Moderation Support Tools:** Worked on NLP-based models to assist content moderation, helping automate the detection of spam, policy violations, and abusive content. This improved moderation response time by 20%, reducing manual workload for moderators.
- **Performed Data Analysis to Support Product and Growth Teams:** Conducted A/B testing and statistical analysis to assess the impact of platform changes on user behavior, providing insights that contributed to a 10% increase in user retention.

Research Assistant in Data Science Lab, *University of Florida* 07/2016 – 05/2017

- Conducted research on probabilistic knowledge base construction, collaborating with Dr. Daisy Zhe Wang to extract structured information from large text corpora using Python, Pandas, and Scikit-learn.
- Developed machine learning pipelines for text preprocessing, entity recognition, and probabilistic reasoning, improving data integration efficiency for large-scale knowledge graphs.

EDUCATION

Master of Science - Mathematics, *University of Florida* 09/2012 – 05/2017

SKILLS

Machine Learning & AI
PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain

Computer Vision
OpenCV, CNNs, Vision Transformers (ViTs)

Data Engineering & Pipelines
Python, C++, Pandas, Airflow, Kafka, Docker, Kubernetes

LLMs & NLP
BERT, GPT-based models, SpaCy, NLTK, OpenAI API

Big Data & Cloud Infrastructure
PySpark, SQL, Snowflake, AWS, GCP, Azure, Domino Data Labs

Analytics & Experimentation
A/B Testing, PowerBI, Tableau, Matplotlib, Seaborn